PHYSICS SIMULATION PROJECT

Pendulum Simulation with Wind & Air Resistance

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COMPUTATIONAL PHYSICS SIMULATION

Project Overview

OBJECTIVE

Simulate the motion of a pendulum under the combined effects of gravity, wind, and air resistance.

ANALYSIS

Investigate how energy is dissipated over time and how external forces alter the system dynamics.

METHOD

Implement a numerical ODE solver (Runge-Kutta) using Python's `scipy.integrate.solve_ivp`.

OUTPUTS

Generate time-series plots, energy analysis, phase-space diagrams, and fitted decay models.

SECOND-ORDER ODE

Mathematical Model

$$\ddot{\theta} = -(g / L) \cdot \sin(\theta) - (b / mL) \cdot \dot{\theta} + (F_{wind} / mL) \cdot \cos(\theta)$$

Gravity Damping Wind Forcing

PARAMETER DEFINITIONS

θ	Angular displacement (rad)
g	Gravitational acceleration (9.81 m/s²)
L	Pendulum length (m)
m	Pendulum mass (kg)
b	Damping coefficient (kg/s)
F_{wind}	Applied wind force (N)

Forces in the System

External forces acting on the pendulum bob

Wind Force

FORCE

$$F_{wind} = F_0$$

(constant horizontal force)

TORQUE CONTRIBUTION

$$\tau_{wind} = F_0 \cdot L \cdot \cos(\theta)$$

A constant horizontal force applied to the pendulum bob. Its angular contribution depends on the instantaneous angle θ, coupling wind effects to the pendulum geometry.

Air Resistance

DRAG FORCE

$$F_{drag} = -b \cdot v = -b \cdot L \cdot \dot{\theta}$$

(velocity-dependent damping)

TORQUE CONTRIBUTION

$$\tau_{drag} = -b \cdot L^2 \cdot \dot{\theta}$$

A velocity-dependent damping force opposing motion. Proportional to angular velocity with damping coefficient b, this force is responsible for energy dissipation.

COMPUTATIONAL METHOD

Numerical Method: Runge-Kutta (RK45)

1 ODE Solver

`solve_ivp`

2 State Vector

The system is expressed as two first-order ODEs: $d\theta/dt = \omega$ and $d\omega/dt = f(\theta, \omega, t)$, forming the state vector **[θ, ω]**.

3 Initial Conditions

The solver accepts initial angle θ_0 and angular velocity ω_0, then computes the full time evolution of the system.

RK45 ITERATION STEPS

k_1 → k_2 → k_3 → k_4

k_1	k_2	k_3	k_4
Slope at start	Slope at midpoint	Corrected midpoint	Slope at end

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)\cdot h$$

4 Implementation

Python call to scipy's adaptive RK45 integrator:

```
sol = solve_ivp(
    pendulum,
    [0, t_max],
    [θ₀, ω₀],
    t_eval=t,
    method="RK45"
)
```

Adaptive step size ensures accuracy while minimizing computation

ENERGY ANALYSIS

Energy Framework

Kinetic Energy

$$KE = \frac{1}{2} \cdot m \cdot (L \cdot \dot{\theta})^2$$

Maximum at the equilibrium position where angular velocity peaks.

Potential Energy

$$PE = m \cdot g \cdot L \cdot (1 - \cos\theta)$$

Maximum at the turning points where the pendulum momentarily stops.

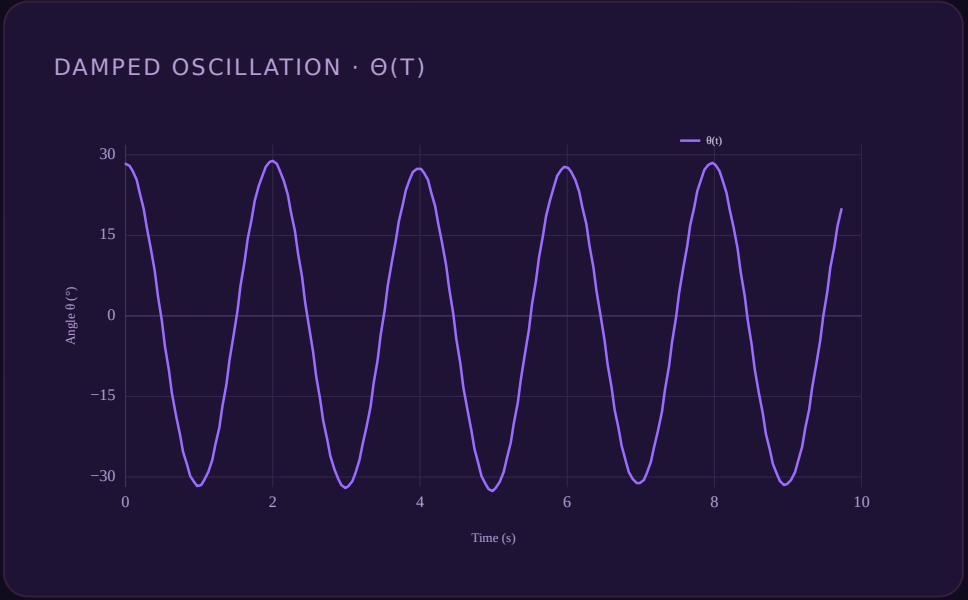
Total Mechanical Energy

$$E_{total} = KE + PE$$

Decreases monotonically over time due to dissipation from air resistance. The decay follows approximately:

$$E(t) \approx E \cdot e$$

Pendulum with Wind and Air Resistance



〰 Angular Dynamics

The pendulum oscillates with progressively decreasing amplitude as energy is dissipated by air resistance. The wind force introduces a slight asymmetry, shifting the equilibrium position.

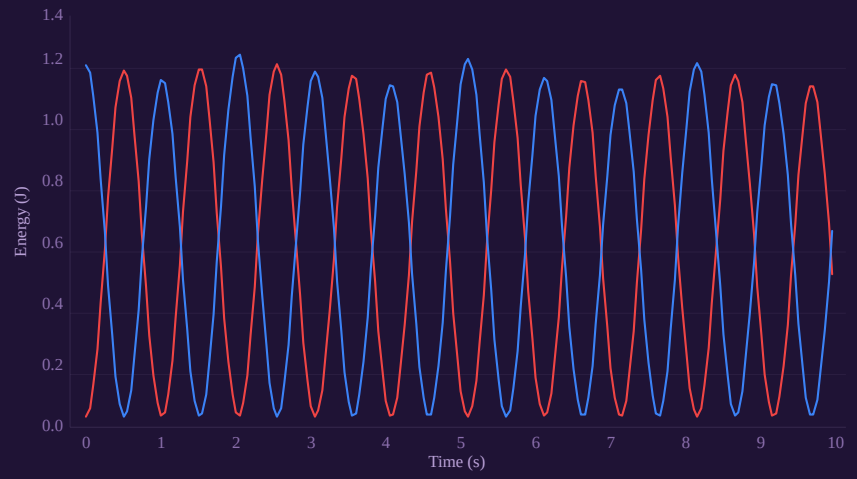
Over time, the system settles toward a new steady state determined by the balance of gravitational and wind forces.

> **Key Observation**
> The exponential envelope $A(t) = A_0 \cdot e^{-\gamma t}$ bounds the oscillation, confirming linear damping behavior.

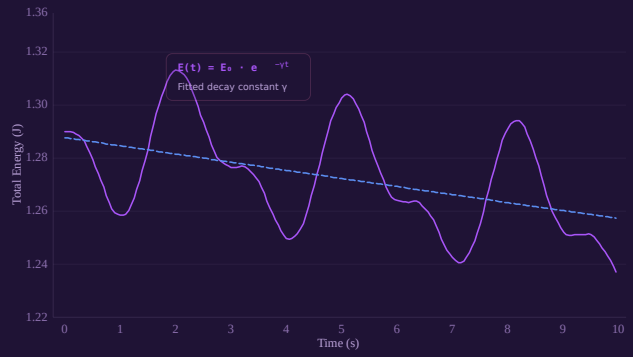
- $\theta_0 = 0.5$ rad Initial angle
- $\omega \approx 2.5$ rad/s Natural freq.
- $\gamma \approx 0.10$ s^{-1} Decay rate

Energy Plots

COMPARING KINETIC AND POTENTIAL ENERGY



ENERGY DECAY WITH FITTED CURVE

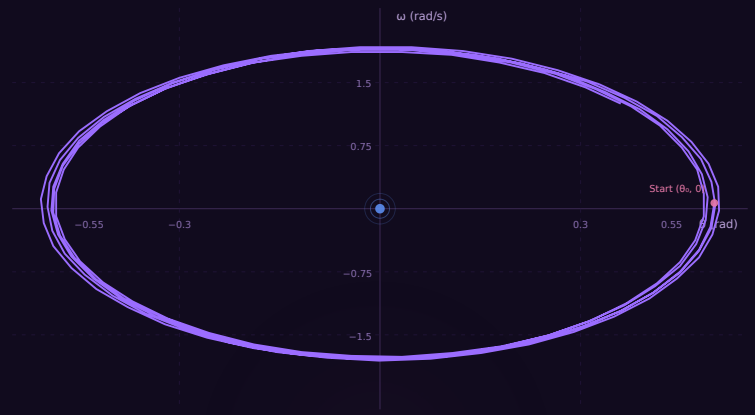


KE and PE oscillate out of phase — energy transfers between kinetic and potential forms each half-cycle. Total mechanical energy decreases monotonically as air resistance dissipates energy, closely following the exponential model $E(t) \approx E_0 \cdot e^{-\gamma t}$.

Fitted Decay & Phase Space

PHASE PORTRAIT

Angular velocity ω vs. angle θ — spiral toward equilibrium



EXPONENTIAL DECAY FIT

$$E(t) = 1.3101 \cdot e^{-0.0041t}$$

An exponential decay model was fitted to the total energy data. The fitted parameters yield $E_0 = 1.3101$ J and decay constant $\gamma = 0.0041\ s^{-1}$, characterizing the rate of energy dissipation due to air resistance.

Model Validation

The close agreement between simulation data and the analytical fit validates the linear damping model used in the system.

Decay Constant $\gamma = 0.0041\ s^{-1}$

Quantifies the timescale of energy loss — higher γ means faster dissipation and quicker settling to equilibrium.

Phase Space Spiral

The inward spiral confirms continuous energy loss — each orbit is smaller as the system asymptotically approaches its steady state.

KEY FINDINGS

Conclusions

1 Damped Dynamics

The simulation accurately models the damped oscillatory motion of a pendulum subject to wind forcing and velocity-dependent air resistance.

2 Numerical Validation

The Runge-Kutta method (RK45) via `solve_ivp` provides a robust and accurate numerical approximation of the nonlinear system behavior.

3 Energy Dissipation

Total mechanical energy decays exponentially over time, consistent with the linear damping model. The oscillation amplitude decreases progressively, leading to smaller and slower swings.

4 Wind Effects

The constant wind force shifts the equilibrium position and introduces asymmetry in the oscillation pattern, demonstrating the coupling between external forcing and pendulum geometry.

Simulation validates theoretical predictions for damped, wind-forced pendulum dynami

Thank You

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Simulation source code and full analysis available on GitH